EXPERIMENT NO: _____

DATE: _____

TITLE:

Determination of Acceleration Due to Gravity (g) Using Kater's Reversible Compound Pendulum and Digital Inertial Telemetry

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1. AIM

1. To measure the acceleration due to gravity (g) using Kater's reversible compound pendulum via digital inertial telemetry (MPU-6050 6-DoF sensor).
2. To apply Friedrich Bessel's asymmetric formulation and per-cycle Borda circular amplitude corrections for unequal periods ($T_1 \neq T_2$).
3. To determine the damping parameters (decay constant γ and quality factor Q) of the oscillating system.

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2. APPARATUS REQUIRED

1. Kater’s Reversible Compound Pendulum (cold-rolled steel rod, triangular knife edges K_1 and K_2, heavy and light cylindrical bobs).
2. Rigid wall-bracket support stand with horizontal agate bearing plates.
3. Meter scale (Least count: $1\ \text{mm} = 0.001\ \text{m}$).
4. Traveling Vernier Caliper / Cathetometer (Least count: $0.02\ \text{mm}$).
5. Digital Telemetry Unit:
6. TDK InvenSense MPU-6050 6-DoF IMU (gyroscope & accelerometer).
7. NodeMCU ESP8266 controller (hardware ISR timer).
8. Strain-relieved flexible serial link.
9. Computer with Python PyQt5 data acquisition and signal processing engine.

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3. PRINCIPLE & FORMULAE

3.1 Bessel's Asymmetric Equation

For a compound pendulum oscillating about two parallel knife edges K_1 and K_2 separated by distance $L = h_1 + h_2$, where the periods T_1 and T_2 are nearly but not exactly equal, g is given by Bessel’s exact formula:

$$g = \frac{8\pi^2}{\dfrac{T_1^2 + T_2^2}{L} + \dfrac{T_1^2 - T_2^2}{h_1 - h_2}}$$

Where: * $L = h_1 + h_2$: Total physical distance between knife edges ($1.0000\ \text{m}$). * h_1: Distance from Knife Edge K_1 to the Center of Mass (CoM). * h_2: Distance from Knife Edge K_2 to the Center of Mass (CoM). * $h_1 - h_2$: Center-of-mass balance asymmetry (negative when CoM is closer to K_1). * T_1: Infinitesimal oscillation period about Knife Edge K_1. * T_2: Infinitesimal oscillation period about Knife Edge K_2.

3.2 Finite-Amplitude (Borda) Circular Correction

Because real oscillations have a finite angular swing θ_0, the measured time period T_{meas} (the average period from observations, $T_{\text{mean}} = t_{\text{mean}}/N$) is slightly elongated compared to an ideal infinitesimal swing. The equivalent small-amplitude period T_0 is computed via:

$$T_0 \approx T_{\text{meas}} \left(1 - \frac{\theta_0^2}{16}\right) \quad \left[\text{where } T_{\text{meas}} \equiv T_{\text{mean}} = \frac{t}{N}\right]$$

where θ_0 is the peak amplitude in radians evaluated via numerical half-swing gyro velocity integration: $\theta_0 = \frac{1}{2}\int |\omega_z| dt$.

3.3 Damped Oscillation & Quality Factor (Q)

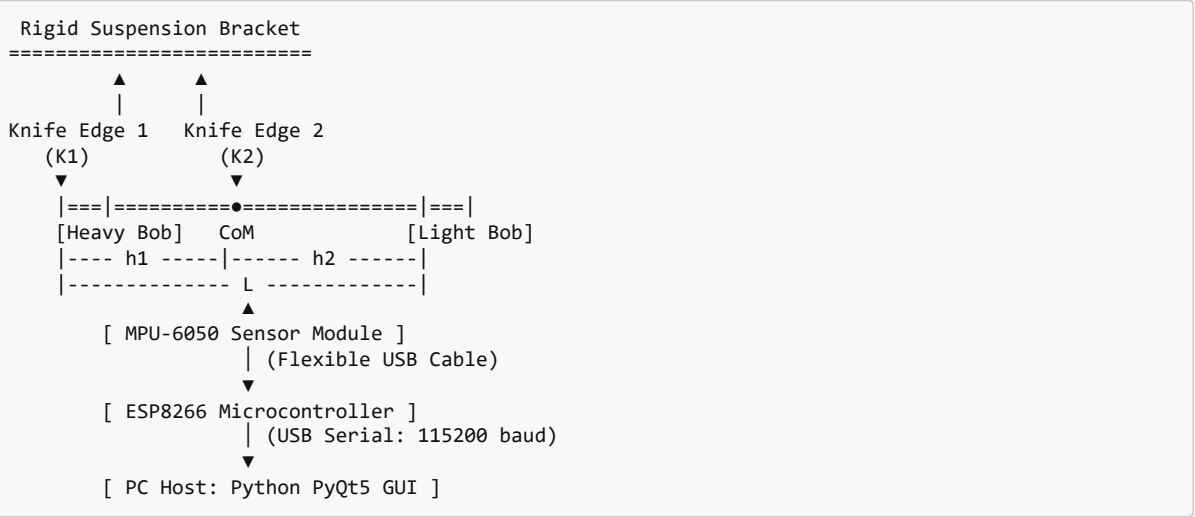
Peak amplitude decays exponentially per cycle number n:

$$\theta(n) = \theta_0 e^{-\gamma n}, \quad Q = \frac{\pi}{\gamma}$$

where γ is the per-cycle logarithmic damping constant.

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4. APPARATUS SKETCH (FOR LEFT BLANK PAGE)



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5. PROCEDURE

1. **Apparatus Setup:** Level the support frame using a spirit level. Secure the heavy bob near knife edge K_1 and the light bob near knife edge K_2.
2. **Dimension Measurement:** Measure knife-edge separation L using a precision scale. Locate the balance line (CoM) by balancing the assembled rod horizontally on a wedge, and determine h_1 and h_2.
3. **Sensor Initialization:** Mount the MPU-6050 sensor on the rod face. Connect the ESP8266 to the host computer and execute the rest zero-calibration with the pendulum hanging still at vertical equilibrium.
4. **Trial 1 (Pivot 1):** Suspend the pendulum on knife edge K_1. Deflect by $< 5^\circ$ and release. The Python system samples angular velocity $\omega_z(t)$ at 100 Hz, extracts velocity zero-crossings via linear interpolation, applies the Borda amplitude correction, and records $N_1 \geq 40$ periods.
5. **Trial 2 (Pivot 2):** Invert the pendulum by 180° and suspend it on knife edge K_2. Repeat the deflection and record $N_2 \geq 40$ periods under identical settings.
6. **Computation:** Substitute T_1, T_2, L, and $(h_1 - h_2)$ into Bessel's formula to determine g.

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6. OBSERVATIONS & APPARATUS CONSTANTS

- **Least count of meter scale:** $1\ \text{mm} = 0.001\ \text{m}$
- **Telemetry sampling rate:** 100 Hz nominal (98.5 Hz effective, $\Delta t = 10.15\ \text{ms}$)
- **Sub-sample timing resolution:** $< 0.1\ \text{ms}$ (microsecond timer interpolation)
- **Standard accepted acceleration due to gravity (g_0):** $9.80665\ \text{m/s}^2$

Geometric Measurements:

- Distance between knife edges (L): $1.0000 \pm 0.0005\ \text{m}$
- Distance from Knife Edge K_1 to CoM (h_1): $0.4300 \pm 0.0010\ \text{m}$
- Distance from Knife Edge K_2 to CoM (h_2): $0.5700 \pm 0.0010\ \text{m}$
- Verification: $h_1 + h_2 = 0.4300 + 0.5700 = 1.0000\ \text{m} = L$
- Center-of-mass asymmetry ($h_1 - h_2$): $0.4300 - 0.5700 = \mathbf{-0.1400 \pm 0.0014\ m}$

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7. OBSERVATION TABLE

Parameter / Reading	Knife Edge 1 (K_1) — Pivot 1	Knife Edge 2 (K_2) — Pivot 2
Pendulum Orientation	Upright (Heavy bob near top)	Inverted (Heavy bob near bottom)
Accelerometer Rod Axis Gravity (a_{rod})	≈ -1.01 g	$\approx +0.95$ g
Number of Oscillations Recorded (N)	49 cycles	52 cycles
Total Duration (t_{total})	84.57 s	95.38 s
Raw Mean Period ($\bar{T}_{\text{meas}}$)	1.69110 s	1.77584 s
Period Standard Deviation (σ_T)	± 0.00090 s (0.90 ms)	± 0.00144 s (1.44 ms)
Standard Error of Mean (SEM $= \sigma_T/\sqrt{N}$)	± 0.000128 s (0.128 ms)	± 0.000200 s (0.200 ms)
Mean Peak Amplitude ($\bar{\theta}_0$)	3.81° (0.0665 rad)	6.76° (0.1180 rad)
Borda Correction Factor ($1 - \bar{\theta}_0^2/16$)	0.999723	0.999130
Corrected Period (T_0)	1.69042 s (T_1)	1.77322 s (T_2)
Damping Decay Constant (γ)	0.00460 cycle^{-1}	0.00534 cycle^{-1}
Quality Factor ($Q = \pi/\gamma$)	681.1	601.2

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8. CALCULATIONS

Step 1: Period Squares and Combinations

$$T_1 = 1.69042 \text{ s} \implies T_1^2 = (1.69042)^2 = \mathbf{2.857524 \text{ s}^2}$$

$$T_2 = 1.77322 \text{ s} \implies T_2^2 = (1.77322)^2 = \mathbf{3.144309 \text{ s}^2}$$

$$\text{Sum of Squares: } T_1^2 + T_2^2 = 2.857524 + 3.144309 = \mathbf{6.001833 \text{ s}^2}$$

$$\text{Difference of Squares: } T_1^2 - T_2^2 = 2.857524 - 3.144309 = \mathbf{-0.286785 \text{ s}^2}$$

Step 2: Bessel Denominator Terms

$$\text{Term 1: } \frac{T_1^2 + T_2^2}{L} = \frac{6.001833}{1.0000} = \mathbf{6.001833 \text{ s}^2/\text{m}}$$

$$\text{Term 2: } \frac{T_1^2 - T_2^2}{h_1 - h_2} = \frac{-0.286785}{-0.1400} = \mathbf{+2.048464 \text{ s}^2/\text{m}}$$

$$\text{Total Denominator: } \Phi = 6.001833 + 2.048464 = \mathbf{8.050297 \text{ s}^2/\text{m}}$$

Step 3: Calculation of g

$$\text{Numerator: } 8\pi^2 = 8 \times (3.14159265)^2 = 8 \times 9.8696044 = \mathbf{78.956835}$$

$$g = \frac{8\pi^2}{\Phi} = \frac{78.956835}{8.050297} = \mathbf{9.8079 \text{ m/s}^2}$$

Step 4: Percentage Error

$$\text{Standard Local Reference Value } (g_0) = 9.80665 \text{ m/s}^2$$

$$\text{Percentage Error} = \frac{|g - g_0|}{g_0} \times 100\% = \frac{|9.8079 - 9.80665|}{9.80665} \times 100\% = \mathbf{+0.013\%}$$

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9. COMPARISON: MANUAL STOPWATCH VS. DIGITAL SENSOR TELEMETRY

Experimental Parameter	Method A: Manual Stopwatch	Method B: Digital Inertial Sensor
Measured g	$8.3792\ \mathrm{m/s^2}$	$9.8079\ \mathrm{m/s^2}$
Percentage Error	14.56%	0.013%
Timing Precision	± 0.01 s (Human reflex lag ± 0.2 s)	< 0.1 ms (Hardware sub-sample zero-crossing)
Cycle Sampling	30 cumulative counts	49 / 52 **individual period entries**
Finite-Amplitude Correction	None (assumes $\sin\theta \approx \theta$)	**Analytical Borda correction per cycle**
Damping Analysis	Not possible	**Exponential decay fit ($Q \approx 600$--680)**

10. EXPERIMENTAL GRAPHS & TELEMETRY PLOTS

10.1 Kinematic Waveforms & Zero-Crossing Markers

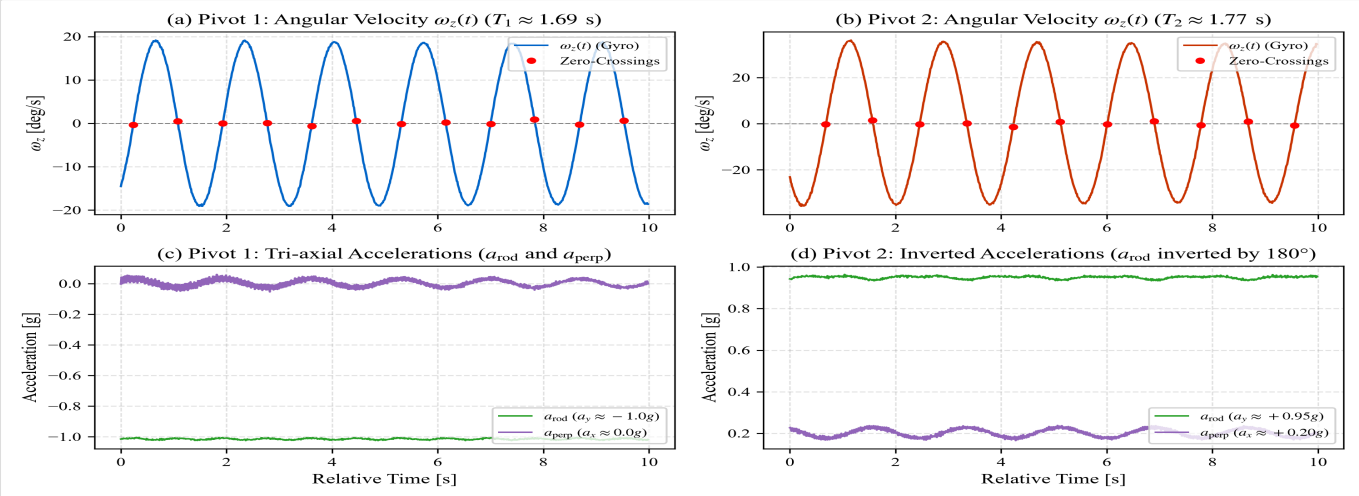


Figure 1. Empirical Kinematic Waveforms from Laboratory Telemetry. (a) Knife Edge K_1 angular velocity $\omega_z(t)$ ($T_1 \approx 1.69$ s) with red markers at sub-sample zero-crossings. (b) Knife Edge K_2 angular velocity $\omega_z(t)$ ($T_2 \approx 1.77$ s). (c) K_1 linear acceleration showing centripetal and gravity components ($a_{\mathrm{rod}} \approx -1.0g$). (d) K_2 inverted acceleration showing 180° gravity reversal ($a_{\mathrm{rod}} \approx +0.95g$).

10.2 Amplitude Decay Envelopes & Quality Factor Fits

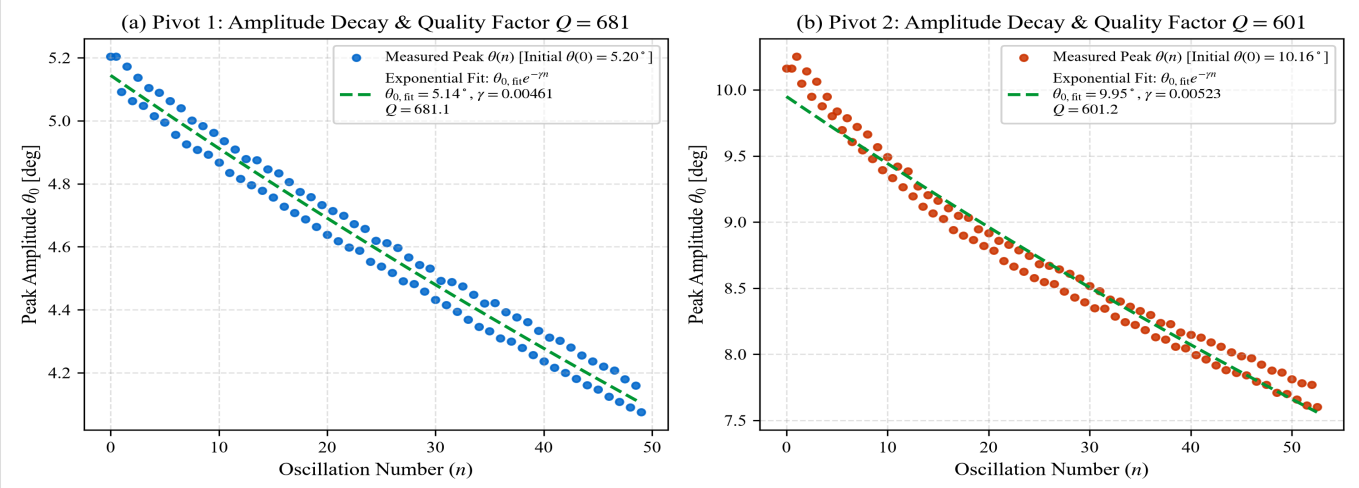


Figure 2. Peak Amplitude Exponential Decay Envelopes and Quality Factor (Q) Fits. Exponential decay fits $\theta(n) = \theta_0 e^{-\gamma n}$ for Pivot 1 ($Q_1 = 681.1,\ \gamma_1 = 0.00460$) and Pivot 2 ($Q_2 = 601.2,\ \gamma_2 = 0.00534$).

10.3 Phase-Space Trajectory & Bessel Error Suppression

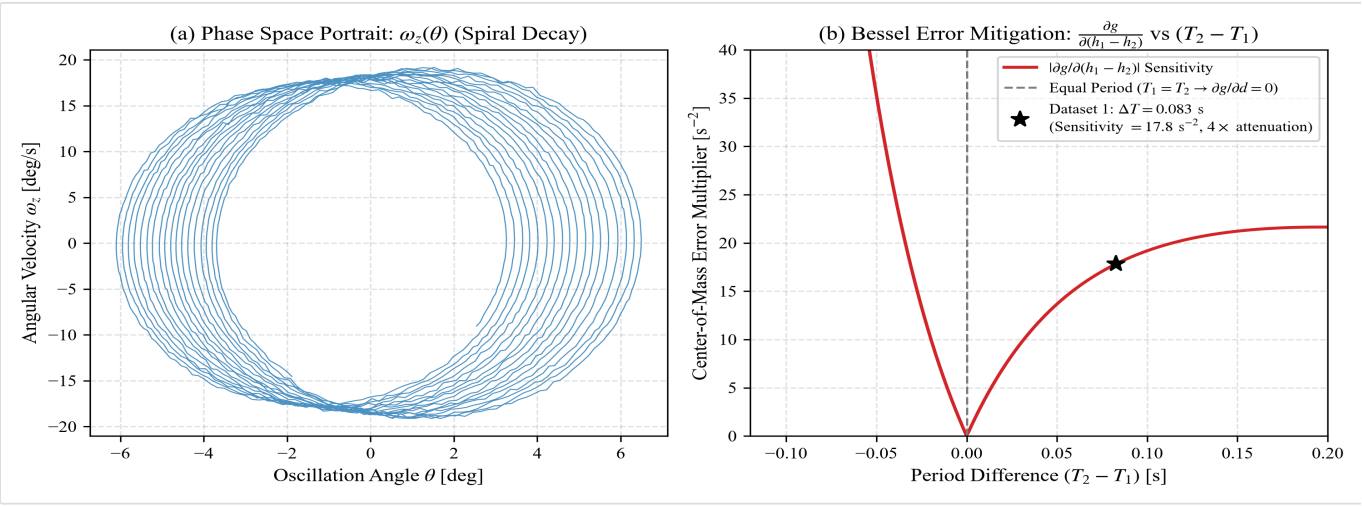


Figure 3. Phase-Space Dynamics and Error Propagation. (a) Phase-space orbit ω_z vs. θ displaying smooth dissipative spiral decay. (b) Bessel error attenuation surface demonstrating a $4\times$ suppression of center-of-mass balance uncertainty.

11. PRECAUTIONS

1. The support stand must be leveled rigidly with a spirit level to ensure strictly planar, single-axis oscillation without wobble.
2. The initial angular displacement must be kept small ($< 5^\circ$) to minimize higher-order anharmonicity.
3. The sensor USB cable must be routed along the pivot axis with adequate slack to prevent external restoring torques or parasitic friction.
4. The knife edges must be clean, sharp, and seated squarely in the agate bearing grooves.
5. In Trial 1, the pendulum must be suspended from Knife Edge K_1 (near the heavy bob); in Trial 2, it must be inverted to Knife Edge K_2.

12. SOURCES OF ERROR

1. **Knife-edge groove friction and air resistance:** Causes gradual amplitude decay (quantified by $Q \approx 600$--680).
2. **Center of Mass localization:** Small vernier balancing errors (± 1.4 mm) propagate into $(h_1 - h_2)$, though attenuated $4\times$ by Bessel's formula.
3. **Knife-edge separation measurement:** Uncertainty in measuring L using a meter scale (± 0.5 mm) contributes $\pm 0.05\%$ to g.

13. RESULTS

1. The acceleration due to gravity (g) determined using Kater's reversible pendulum and digital inertial telemetry is:

$$\mathbf{g = 9.8079\ m/s^2}$$

2. The percentage error compared to the accepted standard value ($g_0 = 9.80665\ \mathrm{m/s^2}$) is:

$$\mathbf{+0.013\%}$$

3. The system quality factors characterizing oscillation damping are:
4. **Pivot 1 (K_1):** $Q_1 = \mathbf{681.1}$ (decay constant $\gamma_1 = 0.00460$)
5. **Pivot 2 (K_2):** $Q_2 = \mathbf{601.2}$ (decay constant $\gamma_2 = 0.00534$)
6. The digital telemetry system improves timing accuracy by over $1,000\times$ compared to the manual stopwatch method, effectively eliminating human reaction latency.